

# **LARA:** Learned Adaptive Recombined Alignments

## A Certifying Foundation Model for Petri Net Trace Alignment

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# The Problem

- **Alignments** explain, event by event, how a trace deviates from a process model
- Optimal alignments need **exact shortest-path search** ( $A^*$ ): often the **bottleneck** on real discovered models
- Purely **learned** aligners are fast, but may be **illegal** or **silently suboptimal**

## The tension

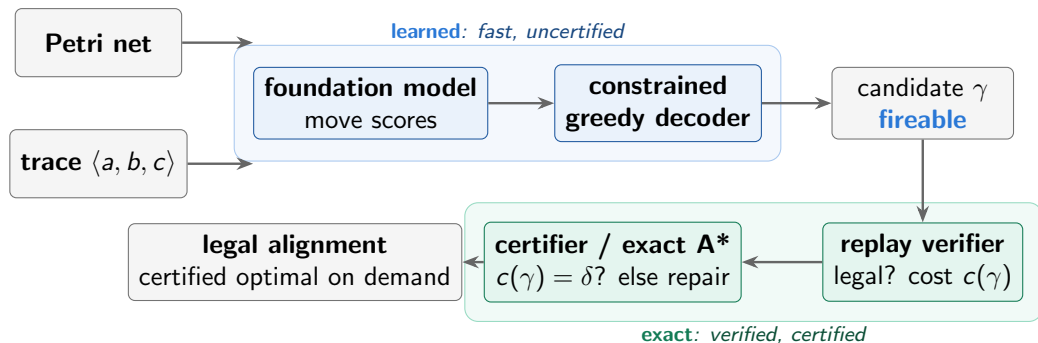
**trust** (exact, slow) vs. **tractability** (learned, fast)

## Fast alignments that are never less trustworthy

A small neural **foundation model**, trained **once** on synthetic data, scores alignment moves for **any Petri net and any log**; exact methods **verify, certify, or repair** every answer.

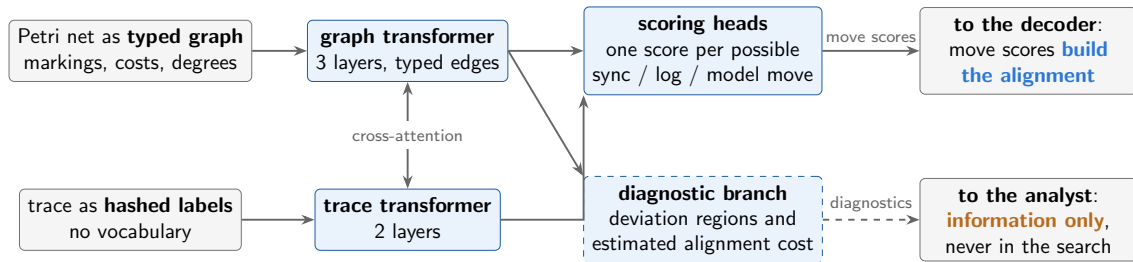
- One invariant: **neural output is never trusted as a certificate**
- Every answer is a **legal alignment with a known cost**; optionally a **certified optimal** one
- **Vocabulary-free**: no retraining, no configuration per log

# The Approach at a Glance



Degrades from **certified optimal** to a **safe upper bound**, **never to wrong**.

# Architecture of the Foundation Model



- Two outputs, two roles: the **move scores** continue down the pipeline and steer the decoder; the **diagnostics** (where the trace deviates, at what estimated cost) go to the **analyst only**
- The diagnostics carry **no guarantee**, so they never influence the search or the result
- Small model: about **3M parameters**, runs on **CPU**; all scores are **trace-dependent**

# Training Loss: Imitating the Exact Aligner

The model learns to **imitate** optimal alignments computed by exact A\*:

$$\mathcal{L} = \underbrace{w_m (\mathcal{L}_{\text{sync}} + \mathcal{L}_{\text{log}} + \mathcal{L}_{\text{model}})}_{\text{imitation of moves, } w_m=1.0} + \underbrace{w_c \mathcal{L}_{\text{cost}} + w_b \mathcal{L}_{\text{bound}} + w_\ell \mathcal{L}_{\text{bal}} + w_e \mathcal{L}_{\text{ent}}}_{\text{diagnostic branch, } w \leq 0.03}$$

- $\mathcal{L}_{\text{sync}}$ : cross-entropy over transition **identities**, not labels: which concrete transition did the optimum fire? This is what teaches **duplicate-label** resolution
- $\mathcal{L}_{\text{log}} / \mathcal{L}_{\text{model}}$ : which events did the optimum **skip**, which transitions did it fire **without a matching event**?
- $\mathcal{L}_{\text{cost}}$ : the summed regional cost bounds must match the **optimal cost**  $\delta$
- $\mathcal{L}_{\text{bound}}, \mathcal{L}_{\text{bal}}, \mathcal{L}_{\text{ent}}$ : regions stay **connected**, **balanced** in size, and **decisive**

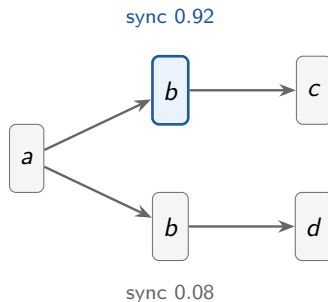
The weights keep **move imitation dominant**; the decomposition is a regularized **diagnostic**, not the primary objective.

# Where Learning Helps: Duplicate Labels

The decoder reads the trace **one event at a time**, always knowing the **current marking** of the Petri net:

- If exactly **one** enabled transition carries the event's label, the choice is **forced by the Petri net**: no learning needed
- If **several** enabled transitions carry the same label, the learned **synchronous score** decides which concrete transition the event refers to
- If **no** such transition is enabled, the **model-move scores** guide a short search that lets the Petri net catch up
- If even that fails, the event is a deviation: it becomes a **log move**

The candidate is always **replayable**; **optimality** is checked by the exact layer.



For the trace  $\langle a, b, c \rangle$ , both transitions labeled  $b$  are enabled; only the **upper** branch can continue with  $c$ , so the learned score selects it.

# Training and Validation Data

**Exact-labeled synthetic pipeline** (imitation of the exact aligner):

- 1 **draw a Petri net** from a curriculum: 80% linear workflows, 20% **duplicate-label** choices differing only in their suffixes
- 2 **perturb** a fitting trace with controlled deviations (rate 0.25: delete, insert, duplicate, swap, append)
- 3 **label** with exact  $A^*$ : optimal alignment **with transition identities**, replay-verified

| Split             | Examples |
|-------------------|----------|
| <b>Training</b>   | 400      |
| <b>Validation</b> | 100      |
| <b>Test</b>       | 100      |

**Hashed labels**  $\Rightarrow$  the model learns **structure**, never vocabulary: one checkpoint for any log.

Training: 22 epochs, about 32 s each, **single CPU**.

# Evaluation: How

**Protocol:** all runs on a **single CPU**. LARA always in **fast mode**: the verified neural candidate is scored **as is**, never repaired, so no exact result contaminates the learned metrics. The **exact A\*** **optimum**  $\delta$  is the ground truth for every sample.

- **In-distribution quality:** 100 held-out examples from the training distribution; measures whether every candidate is **legal** (replayable), how often its cost **equals** the optimum  $\delta$ , and **how far off** it is otherwise
- **Guidance ablation:** identical inputs, but the neural scores are replaced by fixed structural tie-breaks; isolates what **learning** adds over constrained search alone
- **Scaling benchmark:** random block-structured Petri nets with 5 to 40 activities, a family **never seen in training**; locates where the fast path **overtakes exact A\*** in wall time
- **Zero-shot real life:** two public logs (building permits, road traffic fines) with Petri nets discovered by the inductive miner; **unseen vocabularies, no fine-tuning**; tests foundation-model **transfer**

# Evaluation: Results

| In-distribution ( $n = 100$ ) |      |
|-------------------------------|------|
| legal alignments              | 100% |
| certified optimal             | 88%  |
| mean cost gap                 | 0.38 |

Ablation: learning adds **+27.7 points** on the **duplicate-label** family (94.4% vs. 66.7%); zero elsewhere.

- **Legality transfers everywhere:**  
100% on unseen families and real logs
- Fast path **overtakes exact A\*** at about 40 activities ( $3.7\times$ ; 7 to  $12\times$  at 80)
- Real discovered Petri net (invisible-heavy): **18.5 ms** vs. 27.4 ms median
- But: optimal-cost rate **drops to about 30%** on large unseen Petri nets

# What Can Be Improved: Data

- Training Petri nets are **tiny** (3 to 9 transitions): train on the richer **block-structured generator** (loops, concurrency, invisible routing at scale)
- Quality does **not transfer to large Petri nets** (about 30% optimal at 40 activities): **broader training distributions** are the clear next step
- Only **two real-life logs** tested: validate on **larger log collections**

# Takeaway

**Learning proposes, exact reasoning certifies:** the result is **never less trustworthy** than classical alignment.

- **Legality is unconditional:** every produced alignment can be replayed from the initial to the final marking, and this held in **100%** of the cases on synthetic families, on larger unseen Petri nets, and on real-life logs alike
- **88%** of in-distribution candidates already carry the **exact optimal cost**, so one cheap exact call (about 1 ms) certifies them; the other 12% are **repaired exactly**, never returned wrong
- The learned scores pay off exactly on **duplicate labels**: **94.4%** optimal with guidance vs. 66.7% without; everywhere else, constrained search alone is already strong
- The fast path stays at the **forward-pass floor** (8 to 12 ms per trace) and **overtakes exact A\*** at about 40 activities, and already on a real discovered Petri net full of invisible transitions

Checkpoint, scripts, and results: <https://github.com/fit-alessandro-berti/lara-align>

# Appendix

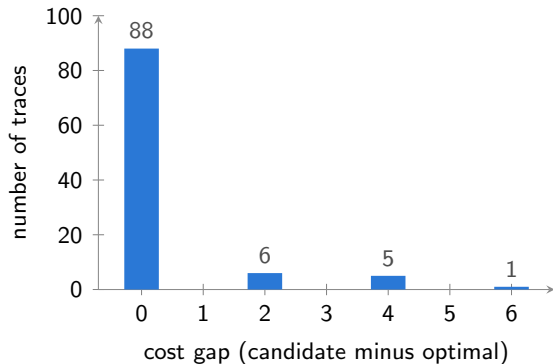
## Detailed Evaluation Results

LARA always in fast mode; exact A\* optimum as ground truth.

## Appendix: In-Distribution Results

| Held-out test split ( $n = 100$ ) |             |
|-----------------------------------|-------------|
| replayable (legal)                | 100.0%      |
| certified optimal                 | 88.0%       |
| exact transition-sequence match   | 68.0%       |
| mean / median cost gap            | 0.38 / 0.00 |
| p95 / max cost gap                | 4.0 / 6.0   |

All non-zero gaps are **even**: the signature of the greedy decoder trading  $k$  synchronous moves for  $k$  model +  $k$  log moves, a penalty of exactly  $2k$ .



## Appendix: Guidance Ablation

Identical inputs and decoder; only the neural scores are replaced by fixed structural tie-breaks:

| Metric ( $n = 100$ )                         | unguided | guided       | $\Delta$     |
|----------------------------------------------|----------|--------------|--------------|
| replayable                                   | 100.0%   | 100.0%       | 0.0          |
| equal to optimum cost                        | 83.0%    | <b>88.0%</b> | +5.0         |
| exact transition-sequence match              | 63.0%    | <b>68.0%</b> | +5.0         |
| mean cost gap                                | 0.48     | <b>0.38</b>  | -0.10        |
| sequence family, optimal ( $n = 82$ )        | 86.6%    | 86.6%        | 0.0          |
| duplicate-label family, optimal ( $n = 18$ ) | 66.7%    | <b>94.4%</b> | <b>+27.7</b> |

- On **sequence** Petri nets every event has at most one matching transition, so guidance changes **nothing**
- The **entire** in-distribution contribution of learning sits in **duplicate-label** identity resolution, exactly as the mechanism predicts

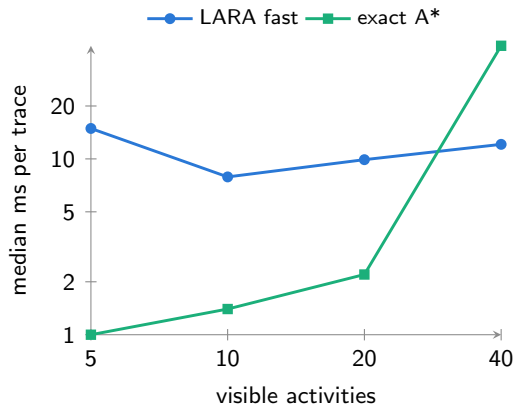
## Appendix: Scaling Benchmark

Random block-structured Petri nets, **never seen in training** (10 samples per configuration, 30 s exact timeout):

| size | legal | optimal | fast med. | exact med.     |
|------|-------|---------|-----------|----------------|
| 5    | 100%  | 80%     | 14.9 ms   | 1.0 ms         |
| 10   | 100%  | 80%     | 7.9 ms    | 1.4 ms         |
| 20   | 100%  | 60%     | 9.9 ms    | 2.2 ms         |
| 40   | 100%  | 30%     | 12.1 ms   | <b>44.0 ms</b> |

(deviation rate 0.15 shown)

**Legality is scale-invariant**; the optimal-cost rate **drops** out of distribution, while the fast path stays at the **forward-pass floor**.



## Appendix: Zero-Shot Results on Real-Life Logs

Same checkpoint, **no fine-tuning**; Petri nets discovered by the inductive miner (noise 0.0 = perfectly fitting, 0.5 = filtered):

| Log          | Noise | Legal | Optimal (trace-wtd.) | Max gap | Median LARA / exact |
|--------------|-------|-------|----------------------|---------|---------------------|
| road traffic | 0.0   | 100%  | 100.0%               | 0       | 8.0 / 1.7 ms        |
| road traffic | 0.5   | 100%  | 94.0%                | 3       | 7.9 / 1.3 ms        |
| receipt      | 0.0   | 100%  | 82.6%                | 5       | 18.5 / 27.4 ms      |
| receipt      | 0.5   | 100%  | 74.6%                | 7       | 8.9 / 8.1 ms        |

- **Legality transfers perfectly** to unseen vocabularies and invisible-heavy discovered models (47 invisible vs. 27 visible transitions on the fitting receipt net)
- Optimality is strong on **frequent** behavior; misses concentrate in **rare, long variants** and remain safe upper bounds
- On the largest discovered Petri net, the fast path **beats exact A\*** on a real log (18.5 vs. 27.4 ms median)